

Accounting and Budgetary Systems for Engineers - INFO7225

MGEN

**Accounting Information System (AIS) and
Business Process Fundamentals (I)**

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CHAPTERS 1 – 7

- 1. Accounting Information Systems (AIS) and the Accountant**
- 2. Accounting on the Internet**
- 3. Information Technology and AIS**
- 4. Accounting and Data Analytics**
- 5. Integrated Accounting and Enterprise Software**
- 6. Internal Control Systems and Risk Management**
- 7. Computer Controls for Organizations and AISs**

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Chapter 1

Accounting Information Systems and the Accountant

After studying this chapter, you will be able to:

1. *Describe* the significant impact of information technology (IT) on the accounting profession and *explain* why you need to study accounting information systems.
2. *Identify* career opportunities that combine accounting and IT knowledge and skills.
3. *Explain* how IT influences accounting systems.
4. *Give examples of* how financial reporting is changing with advances in IT.
5. *Give examples of* how accountants use business intelligence for decision-making.
6. *Identify* what is new in the area of accounting information systems.
7. *Distinguish* between terms such as “systems,” “information systems,” “information technology,” and “accounting information systems.”

Q1-12. Which of the following describes “cloud computing”?

- a. business applications over the Internet
- b. it is a subset of e-business
- c. the ability of a company to exchange data with its subsidiaries
- d. none of the above

Q1-13. How would you describe “big data”?

- a. Cloud computing can be used to support data analytics projects
- b. A combination of large, complex data sets
- c. The definition depends on the capabilities of the organization that manages the data
- d. All of the above

Q1-3. With respect to computerized AISs, computers:

- a. turn data into information
- b. always make audit trails easier to follow
- c. always catch input errors
- d. do not generally process information more quickly than humans

Q1-5. The Sarbanes–Oxley Act of 2002:

- a. enables US officers to wiretap corporate phones if required
- b. was enacted as a reaction to major corporate accounting scandals
- c. allows CPA firms to simultaneously act as management consultants and external auditors for the same organization
- d. does not require that senior executives of public companies take individual responsibility for financial reporting.

Chapter 8

Accounting Information Systems and Business Processes: Part I

8.1 INTRODUCTION

In this chapter and in the next chapter, we discuss the fundamentals of business processes and then focus on several core business processes that are common to most businesses. We begin with a brief refresher of the basics of financial accounting because these concepts are at the heart of an accounting information system (AIS). That is, the fundamental elements of accounting are embedded in automated AISs, and they are the basis for a company's financial statements. These fundamental elements include journals, ledgers, accounts, trial balances, and financial statements. We discuss these concepts from the perspective of simple, paper journals and records because this simplifies the discussion of the key business processes.

Chapter 8

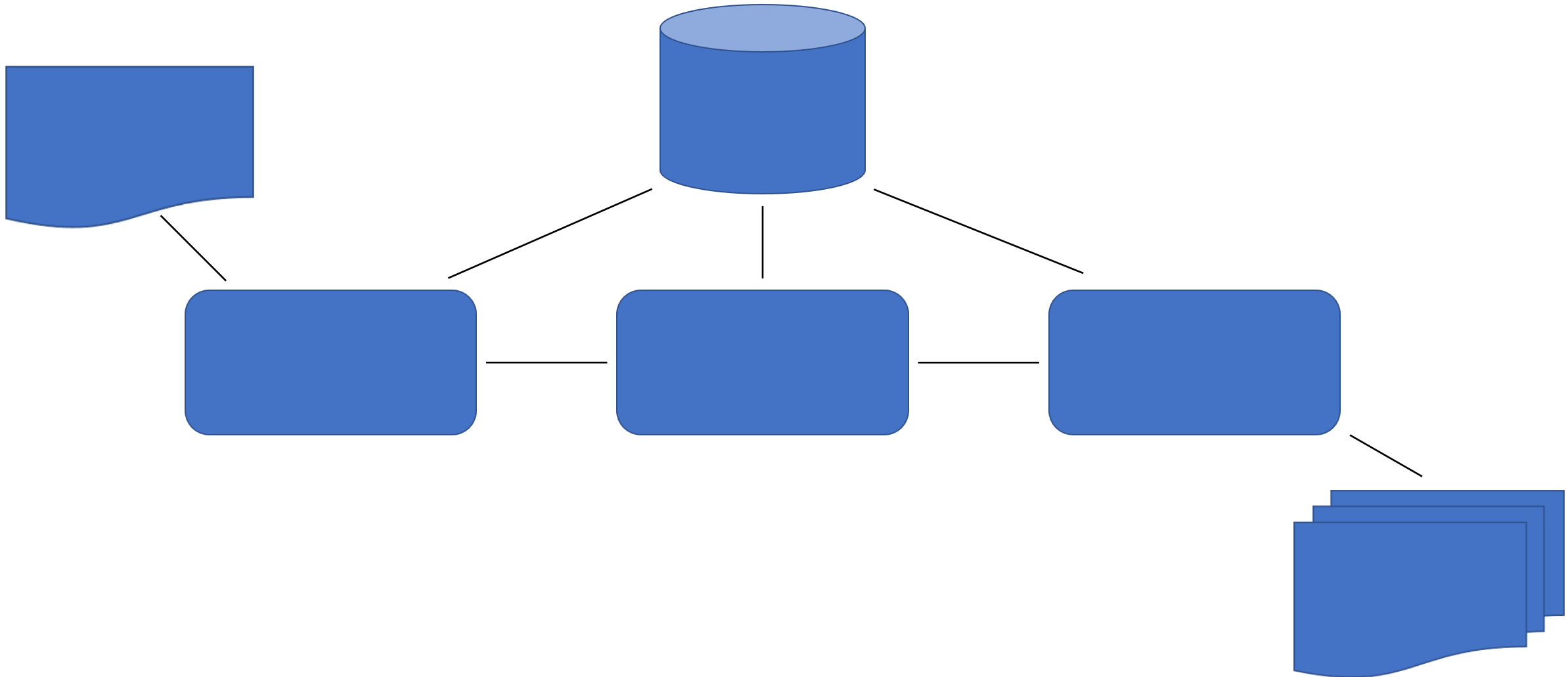
Accounting Information Systems and Business Processes: Part I

After reading this chapter, you will be able to:

1. *List and describe* the steps in the financial accounting process.
2. *Explain and illustrate* the use of journals and ledgers in processing accounting transactions.
3. *Identify* different types of coding systems used by accounting information systems (AISs).
4. *Explain* why planning an AIS begins with the design of outputs.
5. *Describe* the objectives, inputs, and outputs of the sales and purchasing processes.
6. *Explain* why businesses choose outsourcing and offshoring of business processes.

Accounting Information System (AIS):

The key functions performed by an AIS



The Fundamentals of Financial Accounting

- **Accounting records, major classes of accounts, and coding systems of accounts in AIS**
- **The accounting equation**
- **Analyzing transactions using accounting equation**
- **Accounting info flow (8-step accounting cycle: next slide)**

Accounting information flow (8-step accounting cycle)

- 1) Record transaction in a journal.
- 2) Post journal entries to ledgers.
- 3) Prepare an unadjusted trial balance.
- 4) Record & post adjusting journal entries.
- 5) Prepare an adjusted trial balance.
- 6) Prepare financial statements.
- 7) Record & post closing journal entries.
- 8) Prepare a post-closing trial balance.

Special Topics and Summary

- **Special topic 1: cash-basis vs. accrual-basis accounting**
- **Special topic 2: merchandising transactions**
- **Special topic 3: accounting for long-term assets**
- **Summary: generally accepted acct. principles (GAAP)**

The Fundamentals of Financial Accounting:

**Accounting Records:
Document, Journal, and Ledger**

ACCOUNTING RECORDS: DOCUMENT

Source Documents

- Economic events result in the creation of some documents at the beginning (the source) of the transaction.
- These are called source documents and are used to capture and formalize transaction data that the transaction cycle uses for processing.

A sample purchase order

ORDERED BY
BSN BICYCLES
1 Sports Lane
Sports Shop, XX 12345



Purchase
Purchase Order No:
36551

To:
Lu Company
222 Main Street
Pleasantville, XX 23456

Date	Good Through	Account No.		Terms	
3/1/2018	3/30/2018			2/10, n/30	
Item	Description	Quantity	Unit Price	Total	
G124-464	Hot Rider Gloves-Women	15.00	24.95	374.25	
G453-324	Mogul Tire Pumps	20.00	34.95	699.00	

Authorized Signature _____

A sample sales invoice

Voice:
Fax:



Duplicate

Sold To:
BSN BICYCLES
1 Sports Lane
Sports Shop, XX 12345

Ship To:
BSN BICYCLES
1 Sports Lane
Sports Shop, XX 12345

Invoice
Invoice Number:
15563
Invoice Date:
3/5/2018
Page:
1

Customer		Customer PO		Payment Terms	
BSN001		36551		2/10, n/30	
Sales Rep		Shipping Method		Ship Date	Due Date
W. Loman		Rail		3/30/2018	4/2/2018
Quantity	Item	Description		Unit Price	Total
15.00	G124-464	Hot Rider Gloves-Women		\$24.95	374.00
20.00	G453-324	Mogul Tire Pumps		\$34.95	699.00

Check No.

Subtotal \$1,073.25
Sales Tax
Total Invoice Amount \$1,073.25
Payment Received 0.00
TOTAL \$1,073.25

ACCOUNTING RECORDS: DOCUMENT

Source Documents

- Economic events result in the creation of some documents at the beginning (the source) of the transaction.
- These are called source documents and are used to capture and formalize transaction data that the transaction cycle uses for processing.

Product Documents

- They are the result of transaction processing rather than the triggering mechanism for the process.
 - For example, a payroll check to an employee is a product document of the payroll system.

Turnaround Documents

- Product documents of one system that become source documents for another system.

Example: Documents Used and Generated in a Sales Process

Source document:

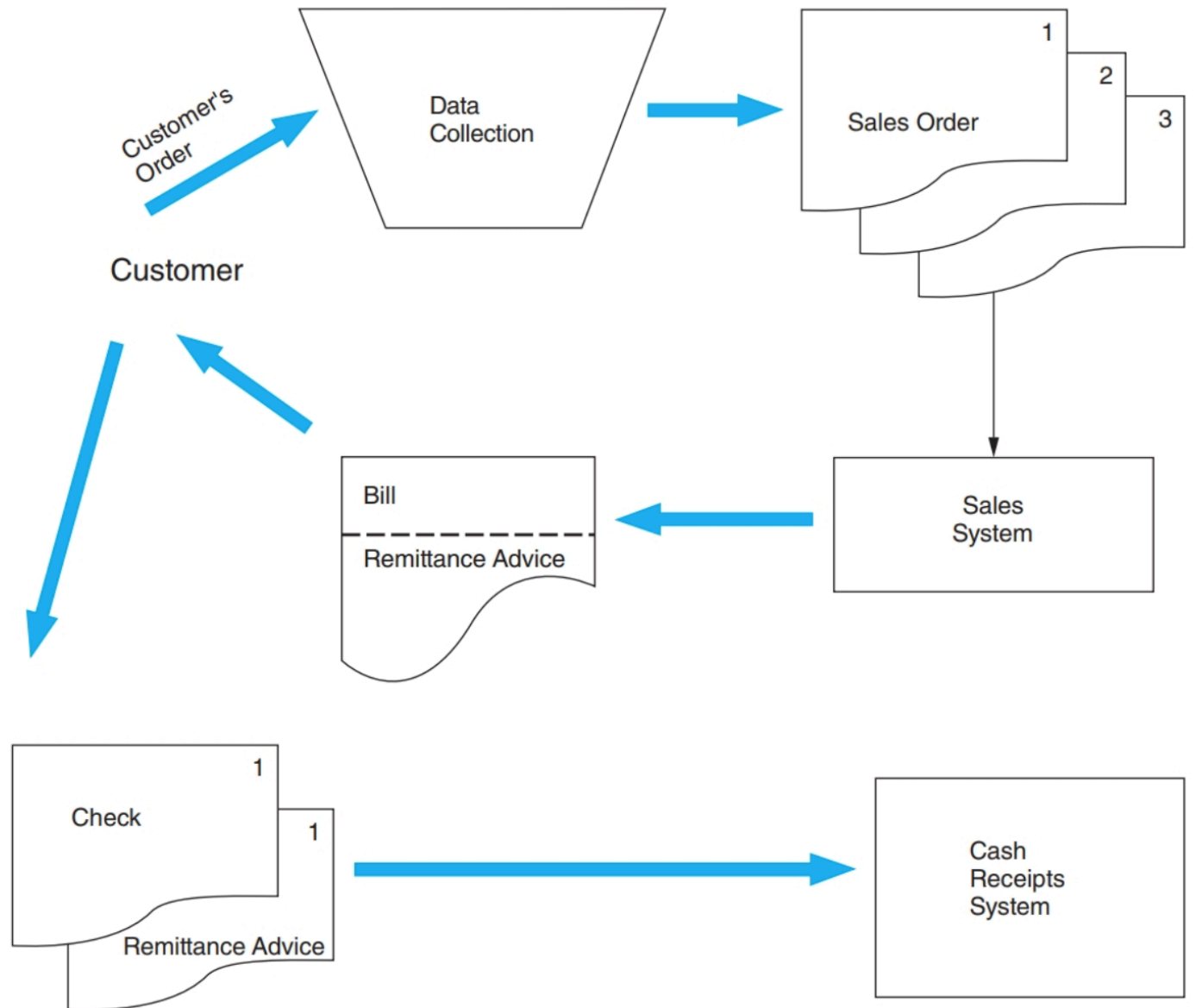
—

Product document:

—

Turnaround document:

—



ACCOUNTING RECORDS: JOURNALS

- A ***journal*** is a chronological record of financial transactions.
 - **Journal entry:**
 - At some point during the transaction process, when all relevant facts about the transaction are known, the event is recorded in a journal in *chronological* order.

- The primary **sources** of data entry into journals are documents.

John Fluno

1764

May 15	To Cash at Fort Stanwix for a shirt	£ 0-16-0
"	To Cash " " "	0-2-0
June 26	To 1. of Chocolate & pair of Shirts	0-10-0
"	To 1 Shirt	0-12-0
"	To 1 pair of breeches	£ 16-0
"	To 2 Sugar	0-3-0
	paid	2 10-0

(a)

28th January	Music Masters Song Book. for Piano	256 pp.	2,000
"	Story & Study Book 2.	194 pp.	500
"	Algebra, Arithmetic Book 2. 2nd.	74 pp.	10,000
"	" " " " " "	128 pp.	500
16th February	Index to School Atlas		500 each
"	March of History Book 1-2		1,500 each
7th February	Index to School Atlas		1,500 each
10th February	Practical English	128 pp.	2,300
"	March of History Book 7	44 pp. 50 pp.	6,000
"	do. " 7a	32 pp. 50 pp.	3,000
11th February	do. " 7	128 pp. 50 pp.	6,000
"	do. " 7B	60 pp. 50 pp.	2,000

(b)

Figure 7.5 Data Storage. (a) General journal and (b) general ledger. (credit a: modification of “Entry in Barent Roseboom’s ledger detailing transactions with John Fluno in 1764” by National Park Service, Public Domain; credit b: modification of “Print Order Book, Holmes McDougall” by Edinburgh City of Print/ Flickr, CC BY 2.0)

TWO CLASSES OF JOURNALS

General Journal

- Firms use the general journal to record nonrecurring, infrequent, and dissimilar transactions.
 - For example, depreciation and closing entries are typically recorded in the general journal.

Special Journal

- A journal used to record a large number of repetitive transactions such as
 - Credit sales, cash receipts, purchases, and cash disbursements

GENERAL JOURNAL													PAGE		
	DATE	DESCRIPTION	POST. REF.	DEBIT					CREDIT						
1	Sept. 1, 20xx	Depreciation Expense	520		5	0	0	0						1	
2		Accumulated Depreciation	210								5	0	0	0	2
3															3
4	Sept. 2, 20xx	Insurance Expense	525		1	2	0	0							4
5		Prepaid Insurance	180								1	2	0	0	5
6															6
7	Sept. 3, 20xx	Cash	101	1	1	0	0	0							7
8		Capital Stock	310							1	1	0	0	0	8
9															9
10															10
11															11
12															12

Special journal example: a sales journal

Date	Customer	Invoice Num.	Acct. Num.	Post	Debit	Credit Sales #401
					Acct. Rec. #102	
Sept. 1	Hewitt Co.	4523	1120		3300	3300
15	Acme Drilling	8821	1298		6825	6825
Oct. 3	Buell Corp.	22987	1030		4000	4000
10	Check Ltd.	66734	1110		8500	8500

ACCOUNTING RECORDS

Documents

- Source documents
- Product documents
- Turnaround documents

Journal (chronological records)

- General journal
- Special journal

Ledger (organized by accounts)

- **General ledger**
 - ▶ A ledger that contains summary-level data for every asset, liability, equity, revenue, and expense account of the organization.
 - ▶ Examples: AR, AP
- **Subsidiary ledger**
 - ▶ A ledger used to record detailed data for a general ledger account with many individual sub-accounts
 - ▶ Breaks down certain general ledger accounts to provide more detailed information to users.



The Fundamentals of Financial Accounting:

Five Classes of Accounts

FIVE CLASSES OF ACCOUNTS

1. Assets
2. Liabilities
3. Equity
4. Revenues
5. Expenses

FIVE CLASSES OF ACCOUNTS: DEFINITION

1. Assets
2. Liabilities
3. Equity
4. Revenues
5. Expenses

What are _____?

- Probable future sacrifice of economic benefits arising from present obligations of a particular entity to transfer assets or provide services to other entities in the future as a result of past transactions or events
 - Current _____: debt or obligation due within one year or, in rare cases, a company's standard operating cycle, whichever is greater
 - Non-current _____: _____ that is expected to be settled in more than one year

FIVE CLASSES OF ACCOUNTS: DEFINITION

1. Assets
2. Liabilities
3. Equity
4. Revenues
5. Expenses

What are _____?

- Tangible or intangible resource owned or controlled by a company, individual, or other entity with the intent that it will provide economic value
 - Current ____ = ____ that will be used or consumed in one year or less
 - Noncurrent ____ = ____ that will be used or consumed over more than one year

FIVE CLASSES OF ACCOUNTS: DEFINITION

1. Assets
2. Liabilities
3. Equity
4. Revenues
5. Expenses

What is _____?

- Residual interest in the assets of an entity that remains after deducting its _____.
 - _____: corporation's primary class of stock issued, with each share representing a partial claim to ownership or a share of the company's business
 - _____: cumulative, undistributed net income or net loss for the business since its inception.

FIVE CLASSES OF ACCOUNTS: DEFINITION

1. Assets
2. Liabilities
3. Equity
4. Revenues
5. Expenses

What are _____?

- Inflows or other enhancements of assets of an entity or settlements of its liabilities (or a combination of both) from delivering or producing goods, rendering services, or other activities that constitute the entity's ongoing major or central operations

What are _____?

- Cost associated with providing goods or services

FIVE CLASSES OF ACCOUNTS: EXAMPLES

1. Assets
2. Liabilities
3. Equity
4. Revenues
5. Expenses

Accounts Payable (AP)		Land	
Accounts Receivable (AR)		Notes Payable	
Bonds Payable		Raw Material Expense	
Building		Retained Earnings	
Cash		Sales	
Common Stock		Service Revenue	
Equipment		Supplies	
Fees Earned		Unearned Revenue	
Prepaid Insurance		Utility	
Interest payable		Wage Payable	
Interest expense		Wage Expense	
Inventory		Warranty Liability	



The Fundamentals of Financial Accounting: Coding System in AIS

A block code used for a company's chart of accounts

Major Accounts		Current Assets Detail	
100–199	Current assets	100	Cash
200–299	Noncurrent assets	110	Marketable securities
300–399	Current liabilities	120	Common stock
400–499	Long-term liabilities	121	Preferred stock
500–599	Owners' equity	122	Bonds
600–699	Revenue	123	Money market certificates
700–799	Cost of goods sold	124	Bank certificates
800–899	Operating expenses	125	Accounts receivable
900–999	Nonoperating income & expenses	130	Prepaid expenses
		140	Inventory
		150	Notes receivable

```
graph LR; A["100–199 Current assets"] --> B["100 Cash"]; A --> C["150 Notes receivable"];
```

Chart of Accounts (1)

ACCOUNT CODE

ACCOUNT NAME

100–199

Current Assets

101	Checking Account
102	Savings Account
103	Petty Cash
120	Accounts Receivable
125	Allowance for Doubtful Accounts
130	Notes Receivable
150	Inventory
160	Supplies
170	Prepaid Rent
180	Prepaid Insurance

200–299

Noncurrent Assets

200	Land
210	Buildings
215	Accumulated Depreciation—Buildings
230	Equipment
235	Accumulated Depreciation—Equipment
240	Furniture & Fixtures
245	Accumulated Depreciation—Furniture & Fixtures
250	Other Assets

ACCOUNT CODE

ACCOUNT NAME

300–399

Liabilities

300	Accounts Payable
310	Wages Payable
321	Employee Income Tax Payable
322	FICA Tax Payable
323	Federal Unemployment Tax Payable
324	State Unemployment Tax Payable
330	Accrued Interest Payable
360	Other Liabilities

400–499

Equity Accounts

400	Common Stock
410	Retained Earnings

Chart of Accounts (2)

ACCOUNT CODE	ACCOUNT NAME
-----------------	--------------

500–599	Revenues
----------------	-----------------

501	Cash Sales	
502	Credit Sales	
510	Sales Returns & Allowances	
511	Sales Discounts	
520	Interest Revenue	its
530	Miscellaneous Revenue	

600–799	Expenses
----------------	-----------------

600	Cost of Goods Sold
611	Wages Expense
612	Commissions Expense
613	Payroll Tax Expense
620	Rent Expense
630	Insurance Expense
640	Supplies Expense
650	Bad Debt Expense
701	Depreciation Expense—Buildings
702	Depreciation Expense—Equipment
703	Depreciation Expense—Furniture & Fixtures
710	Income Tax Expense

900–999	Summary Accounts
----------------	-------------------------

910	Income Summary
-----	----------------

The Fundamentals of Financial Accounting

1. Accounting records, major classes of accounts, and coding systems of accounts in AIS
- 2. Accounting equation**
- 3. Analyzing transactions using accounting equation**
4. Accounting info flow (steps in the accounting cycle)

THE ACCOUNTING EQUATION

(economic resources
owned by an entity)

(creditors' claims
on assets)

(owners' claims
on assets)

$$\begin{array}{l} \text{Cash} + \text{Accounts} + \text{Prepaid} + \text{Equipment} + \text{Truck} \\ \text{Receivable} \quad \text{Insurance} \end{array} = \begin{array}{l} \text{Accounts} + \text{Unearned} + \text{Notes} \\ \text{Payable} \quad \text{Revenue} \quad \text{Payable} \end{array} + \begin{array}{l} \text{Common} + \text{Retained} \\ \text{Stock} \quad \text{Earnings} \end{array}$$

Transactions Summary

(Use the next 2 slides)

1. Issued share capital for \$10,000 cash.
2. Received a bank loan for \$3,000.
3. Purchased equipment for \$3,000 cash.
4. Purchased a truck for \$8,000; paid \$3,000 cash and incurred a bank loan for the balance.
5. Paid \$2,400 for a comprehensive one-year insurance policy effective January 1.
6. Paid \$2,000 cash to reduce the bank loan.
7. Received \$400 as an advance payment for repair services to be provided over the next two months as follows: \$300 for February, \$100 for March.
8. Performed repairs for \$8,000 cash and \$2,000 on credit.
9. Paid a total of \$7,100 for operating expenses incurred during the month; also incurred an expense on account for \$700.
10. Dividends of \$200 were paid in cash to the only shareholder, Bob Baldwin.

Effects on the Accounting Equation

#	Assets	=	Liabilities	+	Equity	Debit	Credit
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Transactions Summary

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Effects on the Accounting Equation

#	Assets	=	Liabilities	+	Equity	Debit	Credit
1							
2							
3							
4							
5							

Transactions Summary

6. Paid \$2,000 cash to reduce the bank loan.
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Effects on the Accounting Equation

#	Assets	=	Liabilities	+	Equity	Debit	Credit
6							
7							
8							
9							
10							

Trans.	ASSETS						=	LIABILITIES			+	EQUITY							
	Cash	+	Acc. Rec.	+	Prepaid Insur.	+	Equip.	+	Truck	=	Acc. Pay.	+	Unearned Revenue	+	Notes Pay.	+	Common Stock	+	Retained Earnings
1.	+10,000																+10,000		
2.	+3,000														+3,000				
3.	-3,000						+3,000												
4.	-3,000								+8,000						+5,000				
5.	-2,400					+2,400													
6.	-2,000														-2,000				
7.	+400													+400					
8.	+8,000			+2,000															+10,000
9.	-7,100											+700							-1,600
10.	-200																		-3,500
																		-2,000	
																		-700	
																		-200	
	3,700		2,000		2,400		3,000		8,000	=	700		400		6,000		10,000		2,000

These numbers are used to prepare the Income Statement.

These numbers are used to prepare the Income Statement.

IN-CLASS EXERCISE 1

- Indicate the net effect (↑ for increase; ↓ for decrease; 0 for no effect) of each of the following transactions on each part of the accounting equation, Assets = Liabilities + Equity.
- For example, for payment of an accounts payable balance, $A (\downarrow) = L (\downarrow) + E (0)$
 - A. Payment of principal balance of note payable $A () = L () + E ()$
 - B. Purchase of supplies for cash $A () = L () + E ()$
 - C. Payment of dividends to stockholders $A () = L () + E ()$
 - D. Issuance of stock for cash $A () = L () + E ()$
 - E. Billing customer for physician services provided $A () = L () + E ()$

In-class Exercise 2. Calculating Financial Statement Components

The following information is taken from the records of Jasper Inc. at January 31, 2015, after its first month of operations. Assume no dividends were declared in January.

Cash	\$33,000	Equipment	\$30,000
Accounts receivable	82,000	Notes payable	15,000
Unused supplies	2,000	Accounts payable	27,000
Land	25,000	Common stock	?
Building	70,000	Net income	40,000

Required:

- Calculate total assets.
- Calculate total liabilities.
- Calculate common stock.
- Calculate retained earnings.
- Calculate total equity.

In-class Exercise 3. Determining Missing Financial Information

Required: Complete the following calculations for each individual company:

- a. If ColorMePink Ltd. has a retained earnings opening balance of \$50,000 at the beginning of the year, and an ending balance of \$40,000 at the end of the year, what would be the net income/loss, if dividends paid were \$20,000?
- b. If ForksAndSpoons Ltd. has net income of \$150,000, dividends paid of \$40,000 and a retained earnings ending balance of \$130,000, what would be the retained earnings opening balance?
- c. If CupsAndSaucers Ltd. has a retained earnings opening balance of \$75,000 at the beginning of the year, and an ending balance of \$40,000 at the end of the year, what would be the dividends paid, if the net loss was \$35,000?

In-class Exercise 4. Equity – What Causes It to Change

	Assets	=	Liabilities	+	Equity	
Balances at April 1, 2015	\$100,000		\$60,000		\$40,000	
					?	Shares issued in April
					?	April net income(loss)
					?	Dividends paid in April
Balances at April 30, 2015	<u>\$180,000</u>	=	<u>\$130,000</u>	+	<u>?</u>	

a	b	c
Equity	Equity	Equity

Required: Using the information provided above, calculate the net income or net loss realized during April under each of the following independent assumptions.

- a. No shares were issued in April and no dividends were paid.
- b. \$50,000 of shares were issued in April and no dividends were paid.
- c. No shares were issued in April and \$4,000 of dividends were paid in April.

The Fundamentals of Financial Accounting

1. Accounting records, major classes of accounts, and coding systems of accounts in AIS
2. Accounting equation
3. Analyzing transactions using accounting equation
- 4. Accounting info flow (8 steps in the accounting cycle)**



Questions?